



PROGRAM SETUP

Choose a galaxy or stars backdrop and delete the default cat



STEPS TO FOLLOW

1. Open scratch.mit.edu and click "Create"
2. Right-click the cat sprite → Delete
3. Click "Choose a Backdrop" (bottom-right)
4. Search "Stars" or "Galaxy" → click the dark starfield
5. Sign in to auto-save your project!

TIP

A dark background makes the yellow stars pop visually. Contrast is a key principle in game design!

DID YOU KNOW?

The Scratch stage is 480×360 pixels.

$x=0$, $y=0$ is the dead centre.



OBJECT PROPERTIES

Add a Bowl sprite, set size to 80, and position it near the bottom



SCRATCH BLOCKS — ON THE BOWL SPRITE

when  clicked

go to x: (0) y: (-150)

set size to (80) %

TIP

y: -150 places the bowl near the bottom. The stage goes from y=-180 (bottom) to y=180 (top).

DID YOU KNOW?

Every object in a game starts with initialization code that sets position, size, and state before the player can interact.



EVENT HANDLING · LOOPS

Left/right arrow keys slide the bowl across the screen; edge stops it



SCRATCH BLOCKS — ON THE BOWL SPRITE

when  clicked

go to x: (0) y: (-150)

forever

if <key [left arrow] pressed?> then

change x by (-10)

if <key [right arrow] pressed?> then

change x by (10)

if on edge, bounce

TIP

Set Bowl's rotation style to "left-right"

in Sprite Info so "if on edge, bounce"

never flips it upside-down.

DID YOU KNOW?

This input-polling pattern inside a forever loop is the game loop — it runs in Unity, Godot, Pygame at 30–60 fps.



OBJECT TEMPLATES · HIDING

Add a Star sprite and hide it on flag click — it will be the clone template



SCRATCH BLOCKS — ON THE STAR SPRITE

STAR: *when*  *clicked*

hide

TIP

Any sprite that uses clones **MUST** hide its original first. Otherwise the "master" copy sits frozen on the stage forever.

DID YOU KNOW?

This template-hide pattern maps directly to a class definition in OOP — the class sits in memory invisibly until you instantiate (clone) it.



CLONE INSTANTIATION

Make the master Star create clones that fall from top to bottom
and delete themselves



SCRATCH BLOCKS — ON THE STAR SPRITE

STAR master: when  clicked

hide

forever

wait (1) seconds

create clone of [myself]

STAR clone: when I start as a clone

show

repeat until <(y position) < (-170)>

change y by (-5)

delete this clone

TIP

"delete this clone" is essential —

without it, thousands of invisible off-screen stars pile up and slow the game to a crawl.



DID YOU KNOW?

"delete this clone" is the game-object lifecycle: create → update → destroy. Every bullet, enemy, and particle follows this same lifecycle.



RANDOM NUMBERS

Make each clone appear at a random x position so stars fall from different spots



SCRATCH BLOCKS — ON THE STAR SPRITE

STAR clone: when I start as a clone

go to x: (pick random (-200) to (200)) y: (175)

show

repeat until <(y position) < (-170)>

change y by (-5)

delete this clone

TIP

Try pick random (-170) to (170) so stars never spawn partially off-screen.

DID YOU KNOW?

Random number generation drives loot tables, enemy spawns, procedural worlds, and weather systems. Randomness makes games feel alive.



COLLISION DETECTION • POLLING

Detect when a falling star touches the Bowl — celebrate with a speech bubble



SCRATCH BLOCKS — ON THE STAR SPRITE

(inside repeat-until loop)

change `y` by `(-5)`

if `<touching [Bowl]?>` then

say [`★ Caught!`] for `(0.5)` secs

delete this clone

TIP

"touching [Bowl]?" polls every frame

— asking "do these two sprites

overlap?" up to 30 times per second.

DID YOU KNOW?

Collision detection is one of the most important algorithms in games. Unity calls these "colliders", Godot calls them "collision shapes".



VARIABLES • STATE

Create a Score variable and add 1 every time a star is caught



SCRATCH BLOCKS — ON THE STAR SPRITE

(inside if <touching [Bowl]?>)

change [Score] by (1)

delete this clone

TIP

Right-click the Score display on the stage and choose "Large readout" for a big satisfying counter!

DID YOU KNOW?

A variable is the most fundamental concept in all programming. Python, JavaScript, Swift — every language uses them.



GAME LOOP · LIFECYCLE

Add Lives — lose one when a star escapes; game ends when
Lives reach zero



SCRATCH BLOCKS — ON THE STAR SPRITE

(after the repeat-until - star escaped)

```
change [Lives] by (-1)
```

```
if <(Lives) = (0)> then
```

```
  say [Game Over! 🗑️] for (2) secs
```

```
  stop [all]
```

```
delete this clone
```

💡 TIP

Show ♥ hearts for lives by right-clicking the Lives display — or display it as "large readout" for impact!

🧠 DID YOU KNOW?

PLAYING → GAME_OVER is a state machine. Traffic lights, login flows, vending machines, and game modes are all state machines.